

Please stick the barcode label here.

- (b) 20 g of steam at 110°C is bubbled through 200 g of milk at 15°C to make frothy milk. Using the result in (a), estimate the temperature of the frothy milk. (2 marks)

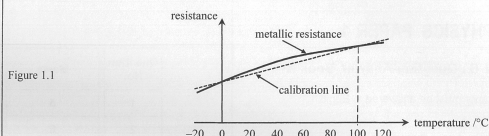
- (c) Would the actual temperature of frothy milk be higher than, equal to or lower than the value found in (b)? Explain. (2 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Section B: Answer ALL questions. Parts marked with * involve knowledge of the extension component. Write your answers in the spaces provided.

1. The solid curve in Figure 1.1 shows how the resistance of a metallic resistance thermometer varies with temperature. This thermometer is calibrated at standard atmospheric pressure for the melting point of ice and the steam point of boiling water. The dotted calibration line represents how the resistance of the thermometer varies with temperature if a linear resistance-temperature relationship is assumed. The deviation of the curve from linearity has been exaggerated in the figure.



- (a) (i) Using the resistances at the calibration points tabulated below, calculate the expected resistance at 60 °C if the resistance varies linearly with temperature. (2 marks)

temperature / °C	resistance / Ω
0	102.00
100	140.51

- (ii) Now if the resistance of the resistance thermometer is the value found in (a)(i), is the actual temperature higher than, lower than or equal to 60 °C? (1 mark)

- (b) In an experiment to determine the specific heat capacity of water c_W , Peter used this calibrated resistance thermometer to measure the temperature of water being heated from 0 °C to 60 °C. Heating was stopped when this thermometer's resistance reached the value found in (a)(i). Assuming negligible heat exchange with the surroundings, no error in measuring the energy supplied and the mass of water, explain whether the experimental value of c_W found is higher than, lower than or the same as the actual value. (2 marks)

Answers written in the margins will not be marked.

Please stick the barcode label here.

- *2. The aqua-lung (a cylinder containing compressed air) for divers has a capacity of $1.0 \times 10^4 \text{ cm}^3$. When the aqua-lung is filled, the air inside has a pressure of 210 atm (atmospheric pressure) at 24 °C. The air in the aqua-lung is allowed to expand through a pressure-reducing valve until its pressure equals that of the surrounding water before it is supplied to divers. Assume that the temperature of the air inside the aqua-lung is always equal to that of the surrounding water.

- (a) A diver stays in water of temperature 24 °C and pressure 2.0 atm at a depth of 10 m. Find the total volume of air (in cm^3) available for the diver from the aqua-lung at this water pressure. (2 marks)

- (b) The supply of air in (a) is sufficient for the diver to remain at such a depth for 1 hour.

- (i) If the diver breathed in the same volume V_0 (in cm^3) of air per minute, find V_0 . (1 mark)

- (ii) If the diver dives deeper where the water is of temperature 20 °C and pressure 4.5 atm, estimate how long (in minutes) the air in a fully-filled aqua-lung would last. Assume that the diver breathes in the same volume of air per minute as that found in (b)(i). (3 marks)

Answers written in the margins will not be marked.

Go on to the next page

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

2016-DSE-PHY 1B-3

3

Go on to the next page

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

mass of gas at constant temperature. The volume V of the gas trapped is read directly from the syringe and the corresponding pressure p is measured by a data-logger via a pressure sensor.

- (a) The initial volume and pressure of the gas are $6.0 \times 10^{-5} \text{ m}^3$ and $1.0 \times 10^5 \text{ Pa}$ respectively at a room temperature of 25°C . Estimate the number of gas molecules trapped in the syringe. (3 marks)

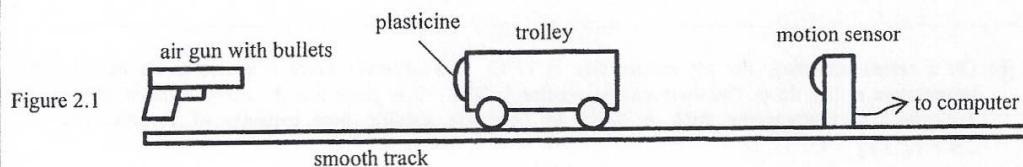
Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

2. The following experimental items are provided to set up an experiment to estimate the speed of a bullet fired from an air gun.

a smooth track
 a trolley
 a motion sensor used to measure the speed of the trolley
 some plasticine
 an air gun and bullets
 an electronic balance

The set-up is shown in Figure 2.1.



Describe the procedures of the experiment. State the physical quantities to be measured and an equation for finding the speed of the bullet. Write down **ONE** precaution for getting a more accurate result. (5 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

- (b) Assuming the power estimated in (a) remains unchanged, determine the flow rate of tap water into the heater, in kg per minute, such that hot water at a temperature of 40°C is delivered by the heater. (2 marks)

- (a) Estimate the power supplied to the tap water by the heater. On a certain day in winter, the temperature of tap water is 15°C . When the heater is in use, it delivers 6 kg of hot water at a temperature of 50°C in 1 minute . Assume that there is no heat exchange between the heater and the surroundings. Given: specific heat capacity of water $= 4200\text{ J kg}^{-1}\text{ }^\circ\text{C}^{-1}$. (3 marks)

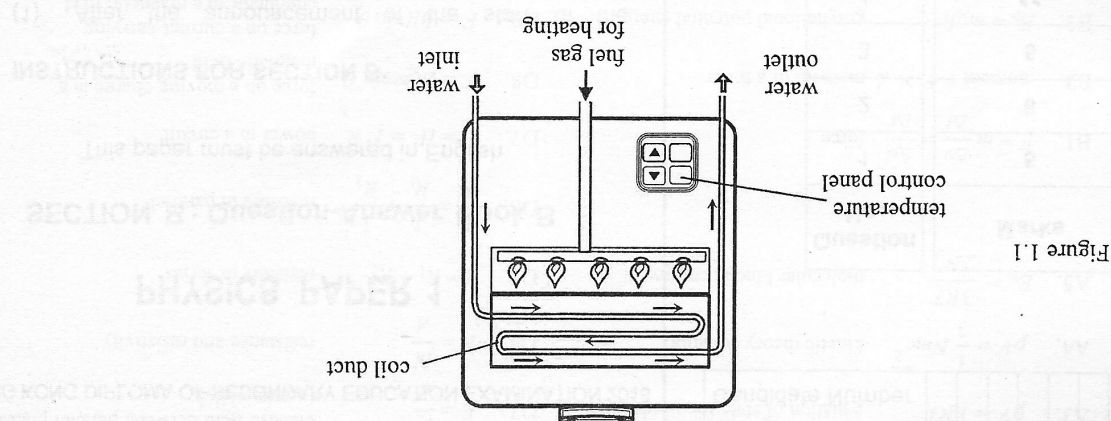


Figure 1.1

1. Figure 1.1 shows a domestic water heater. Tap water entering the heater is heated when it passes through the coil duct and exits as hot water at a certain temperature. Section B: Answer ALL questions. Parts marked with * involve knowledge of the extension component. Write your answers in the spaces provided.

Answers written in the margins will not be marked.

Please stick the barcode label here.

- (b) Some ice cream at $-10\text{ }^{\circ}\text{C}$ is put into a 'thermal bag', of which the inner layer is made of polyethylene foam coated with aluminium foil. The bag is also equipped with a zipper at the top.



The thermal bag is then brought outdoors on a hot sunny day.

- (i) Referring to the heat transfer processes, explain **ONE** feature of this bag that helps keep the ice cream at a low temperature. (1 mark)

.....

.....

.....

- (ii) Suggest **ONE** modification to this bag that would enhance its ability to keep things stored inside at a low temperature. (1 mark)

.....

.....

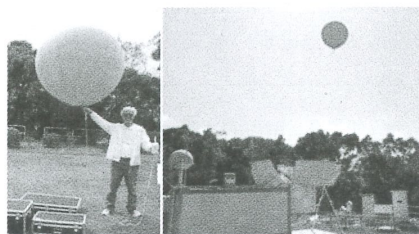
.....

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

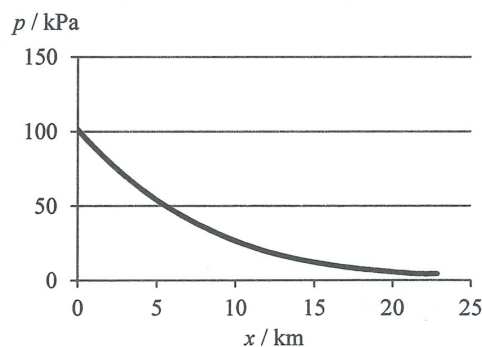
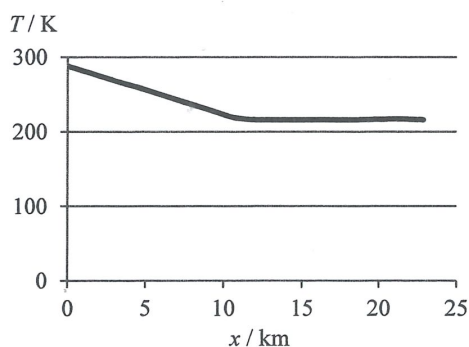
- *2. A weather balloon of volume 0.52 m^3 is filled with helium gas of temperature 15°C and pressure 100 kPa at ground level.



- (a) Find the amount of helium gas (in mol) in the balloon.

(2 marks)

- (b) The following graphs show the variation of air temperature T and atmospheric pressure p with height x above ground level.



The weather balloon is released and rises to the upper atmosphere. Assume that the temperature and pressure of the helium gas in the balloon are the same as those of the air outside at any height x .

- (i) A student believes that as the air temperature decreases in the first 10 km, the volume of the balloon decreases. Referring to the graphs above, explain qualitatively why this belief is not correct. (2 marks)

Answers written in the margins will not be marked.

Please stick the barcode label here.

- (ii) A student used the following method to find the heater's operating power P : remove the heater from the container and record the temperature of the 16 kg of soup after 10 minutes. It is found that the temperature has dropped 9°C . Estimate P . (3 marks)

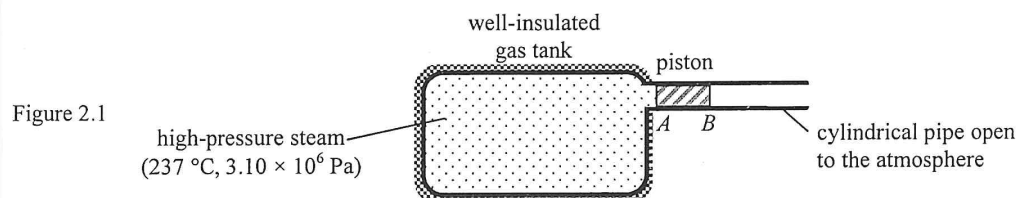
Answers written in the margins will not be marked.

- (iii) If the student repeats the measurement after another 10 minutes, would the corresponding temperature drop be larger than, equal to or smaller than 9°C ? Explain. (2 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

2. Figure 2.1 shows a large gas tank connected with a cylindrical pipe open to the atmosphere. The pipe is fitted with a smooth piston AB . This well-insulated gas tank is filled with high-pressure steam at a temperature of $237\text{ }^{\circ}\text{C}$ under a pressure of $3.10 \times 10^6\text{ Pa}$ while the movable piston is held stationary by a force F_p .
Given: atmospheric pressure = $1.0 \times 10^5\text{ Pa}$



- (a) (i) On Figure 2.1 indicate the force F_p . (1 mark)

- *(ii) By considering the force acting on the piston due to the difference in pressure, find the value of F_p .
The piston has a cross-sectional area of 0.67 m^2 . (2 marks)

Answers written in the margins will not be marked.

- *(iii) Estimate the volume of the gas tank which contains 570 kg steam. You may treat the steam as an ideal gas. Given: mass of one mole of steam = 0.018 kg. (3 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Please stick the barcode label here.

(b) Explain why all the curves become less steep as time elapses.

(2 marks)

.....

.....

.....

.....

.....

(c) (i) On Figure 1.2, indicate the respective curves that correspond to the results of **cup Y** and **cup Z**. (1 mark)

(ii) Explain your answer by referring respectively to the main heat transfer process.

(3 marks)

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

(d) Suggest a material for making cups so that heat loss via conduction can be reduced.

(1 mark)

.....

.....

.....

.....

.....

.....

.....

.....

.....

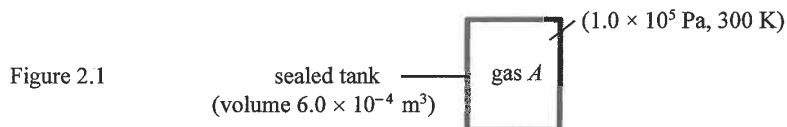
.....

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

- *2. (a) (i) Figure 2.1 shows a sealed tank of volume $6.0 \times 10^{-4} \text{ m}^3$ containing a monatomic gas A at a pressure of $1.0 \times 10^5 \text{ Pa}$ and at a temperature of 300 K .



- (I) Estimate the number of gas molecules in the tank, N . (2 marks)

.....

.....

.....

.....

.....

- (II) Estimate the average kinetic energy of the gas molecules, E_K . (2 marks)

.....

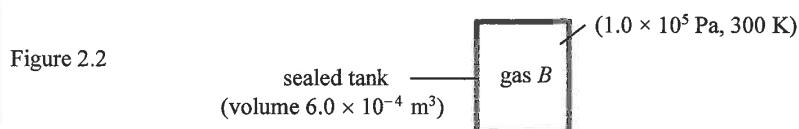
.....

.....

.....

.....

- (ii) Figure 2.2 shows another identical tank containing monatomic gas B under the same pressure and temperature. A molecule of gas B has $\frac{1}{5}$ the mass of a molecule of gas A .



- (I) State whether N and E_K of gas B are larger than, smaller than or the same as the corresponding values of those of gas A found in (a)(i). (2 marks)

.....

.....

.....

Answers written in the margins will not be marked.

Please stick the barcode label here.

- (d) (i) A mercury-in-glass thermometer is not suitable for measuring the temperature of the heated iron cube **directly**. Why? (1 mark)

.....

.....

.....

- (ii) Suggest a kind of thermometer that can directly measure the temperature of this heated cube. (1 mark)

.....

.....

.....

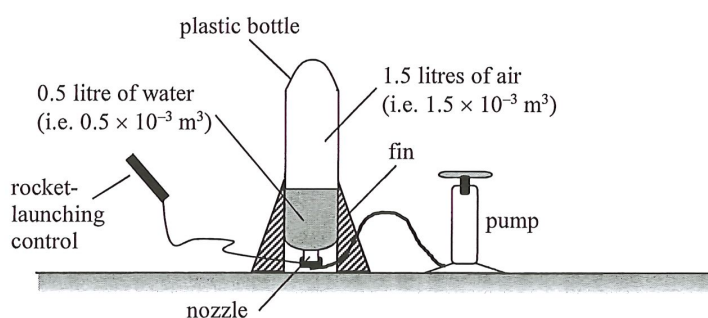
Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

- *2. Figure 2.1 shows a ‘water rocket’ built from a 2-litre plastic bottle with fins attached. The bottle is filled with water such that 1.5 litres of air is trapped inside.

Figure 2.1



Initially the trapped air of temperature 27°C is at atmospheric pressure.

Given: atmospheric pressure = $1.0 \times 10^5 \text{ Pa}$

density of air = 1.20 kg m^{-3}

- (a) Find the mass of the trapped air m_0 . (2 marks)

Air is then pumped slowly into the rocket until the total mass of trapped air inside the bottle becomes $4m_0$. Assume that the trapped air is kept at a temperature of 27°C (i.e. 300 K) and its volume remains constant.

- (b) (i) Find the pressure of the trapped air. (2 marks)

- (ii) Use kinetic theory to explain why the pressure of the trapped air increases. (2 marks)

Answers written in the margins will not be marked.



Section B: Answer ALL questions. Parts marked with * involve knowledge of the extension component. Write your answers in the spaces provided. Take $g = 9.81 \text{ m s}^{-2}$.

1. A bottle of water of temperature 4°C is taken out from a refrigerator and placed on a table. The room temperature is 30°C .



Figure 1.1

- (a) State one way of heat transfer from the surroundings to the bottle of water. (1 mark)

- (b) Explain whether wrapping the bottle with an aluminum foil will speed up the rate of temperature increase. (2 marks)

- (c) After a period of time, water droplets form on the surface of the bottle.

- (i) State the source of the water droplets. (1 mark)

- (ii) Suppose 0.40 g of water droplets are collected on the surface of the bottle. Estimate the latent heat released from these droplets. (2 marks)

Given: specific latent heat of vaporization of water $= 2.26 \times 10^6 \text{ J kg}^{-1}$

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Please stick the barcode label here.

- *4. A sphere of radius 0.50 m rotates about a vertical axis ACB with a constant angular speed of 6 rad s^{-1} , with C being the sphere's centre. A small plasticine P of mass 0.020 kg is stuck on the surface of the sphere such that it rotates on a horizontal plane containing C as shown in Figure 4.1(a).

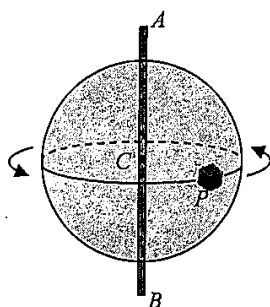


Figure 4.1(a)

Top view from A

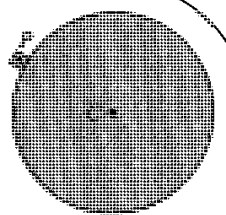


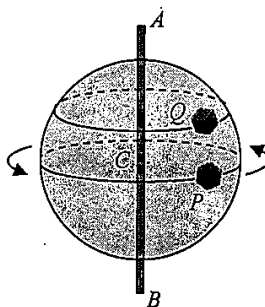
Figure 4.1(b)

- (a) (i) Find the magnitude of the centripetal force acting on P . (2 marks)

- (ii) The sphere's angular speed is increasing. If P finally leaves the sphere at the position shown in Figure 4.1(b) (Top view from A). Indicate on Figure 4.1(b) the direction of motion of P when it just leaves the sphere. (1 mark)

- (b) Another plasticine Q is stuck on the sphere's surface as shown in Figure 4.2.

Figure 4.2



- (i) State whether P and Q have the same angular speed. (1 mark)

- (ii) Explain whether P and Q have the same magnitude of centripetal acceleration. (2 marks)

Answers written in the margins will not be marked.

- *8. A rectangular coil, with N turns and each turn of area $9.00 \times 10^{-4} \text{ m}^2$, is placed in a uniform magnetic field of strength B . Initially the plane containing the coil is parallel to the magnetic field. The coil is then rotated steadily about the axis XY as shown in Figure 8.1(a).

Figure 8.1(a)

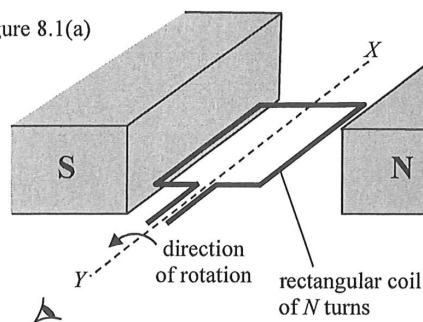
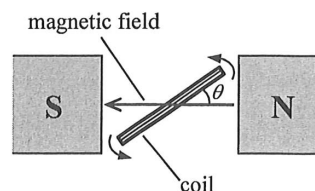
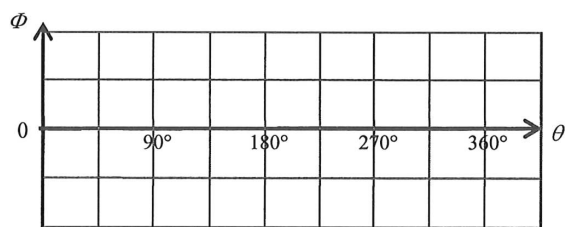


Figure 8.1(b)



- (a) Sketch the variation of magnetic flux Φ passing through each turn of the coil when the coil rotates from $\theta = 0^\circ$ to 360° (see Figure 8.1(b)). (2 marks)



- (b) The maximum magnetic flux through each turn of the coil is $2.70 \times 10^{-4} \text{ Wb}$.

- (i) Determine B (unit: T).

(2 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

- (ii) Calculate the average induced e.m.f. ε in the coil when it is rotated from $\theta = 0^\circ$ to 90° in 0.01 s. Given that the number of turns N is 100. (2 marks)

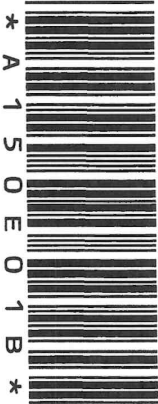
Answers written in the margins will not be marked.

This paper must be answered in English

INSTRUCTIONS FOR SECTION B

- (1) After the announcement of the start of the examination, you should first write your Candidate Number in the space provided on Page 1 and stick barcode labels in the spaces provided on Pages 1, 3, 5, 7 and 9.
- (2) Refer to the general instructions on the cover of the Question Paper for Section A.
- (3) Answer **ALL** questions.
- (4) Write your answers in the spaces provided in this Question-Answer Book. Do not write in the margins. Answers written in the margins will not be marked.
- (5) Graph paper and supplementary answer sheets will be provided on request. Write your Candidate Number, mark the question number box and stick a barcode label on each sheet, and fasten them with string **INSIDE** this Question-Answer Book.
- (6) No extra time will be given to candidates for sticking on the barcode labels or filling in the question number boxes after the 'Time is up' announcement.

1	5
2	6
3	5
4	5
5	7
6	5
7	5
8	7
9	6
Long Questions	
Question No.	
10	11
11	13
12	9
Marks	



Answers written in the margins will not be marked.

(b) The power output of the heater is 2 kW. Estimate the time required.

(2 marks)

(c) Explain why the temperature of water remains at the boiling point after the water starts to boil, even when the heater is still on. (1 mark)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.



(a) The bulb works at its rated value after R becomes stable. Estimate the rated power of the bulb. (2 marks)

(b) Explain why R initially increases and becomes stable after a period of time. (3 marks)

(c) The filament of the bulb has a length of 0.56 m and a uniform cross-section of diameter 0.022 mm. Find the resistivity of the filament at $t = 0$ s. (2 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

(1 mark)

- (b) Estimate the maximum magnetic flux linkage produced by the wire through the search coil. (3 marks)

- (c) A student claims that the search coil can be used to measure the resultant magnetic field produced by the Earth and the wire. Explain briefly whether this claim is correct. (2 marks)

Answers written in the margins will not be marked.



Answers written in the margins will not be marked.

B

HONG KONG EXAMINATIONS AND ASSESSMENT AUTHORITY
HONG KONG DIPLOMA OF SECONDARY EDUCATION EXAMINATION

PHYSICS PAPER 1 (Sample Paper)

Section B : Question-Answer Book B

This paper must be answered in English

INSTRUCTIONS

- (1) Write your Candidate Number in the space provided on Page 1.
- (2) Stick barcode labels in the spaces provided on Pages 1, 3, 5, 7 and 9.
- (3) This section carries 84 marks. Answer **ALL** questions.
- (4) Write your answers in the spaces provided in this Question-Answer Book. Do not write in the margins. Answers written in the margins will not be marked.
- (5) Supplementary answer sheets will be provided on request. Write your Candidate Number, mark the question number box and stick a barcode label on each sheet. Tie them loosely but securely with a string **INSIDE** this Question-Answer Book.
- (6) The diagrams in this section are **NOT** necessarily drawn to scale.

Please stick the barcode label here.

Candidate Number							
------------------	--	--	--	--	--	--	--

	Marker's Use Only	Examiner's Use Only
	Marker No.	Examiner No.
Question No.	Marks	Marks
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		
12		
13		
14		
Total		